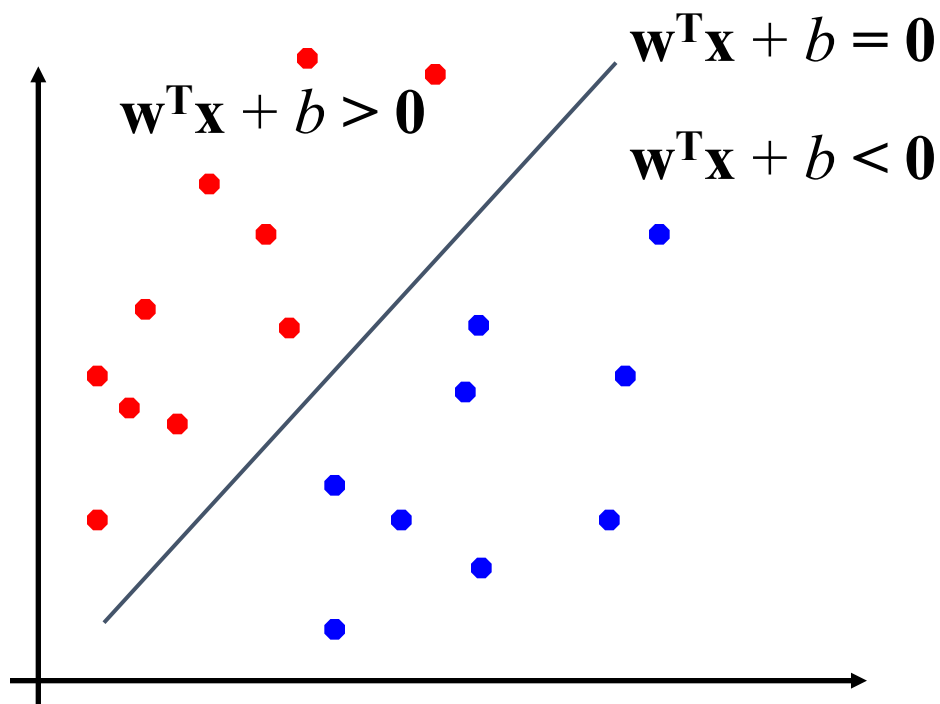


Support Vector Machine

Prof. Arun Chauhan, CSE
GEU, Dehradun

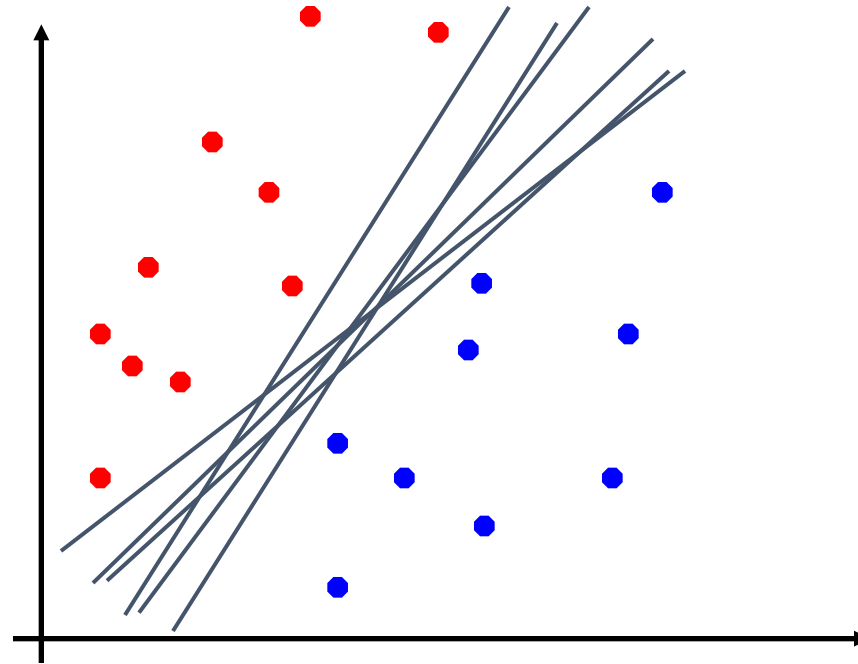
Logistic Regression Revisited: Linear Separators

- Binary classification can be viewed as the task of separating classes in feature space:



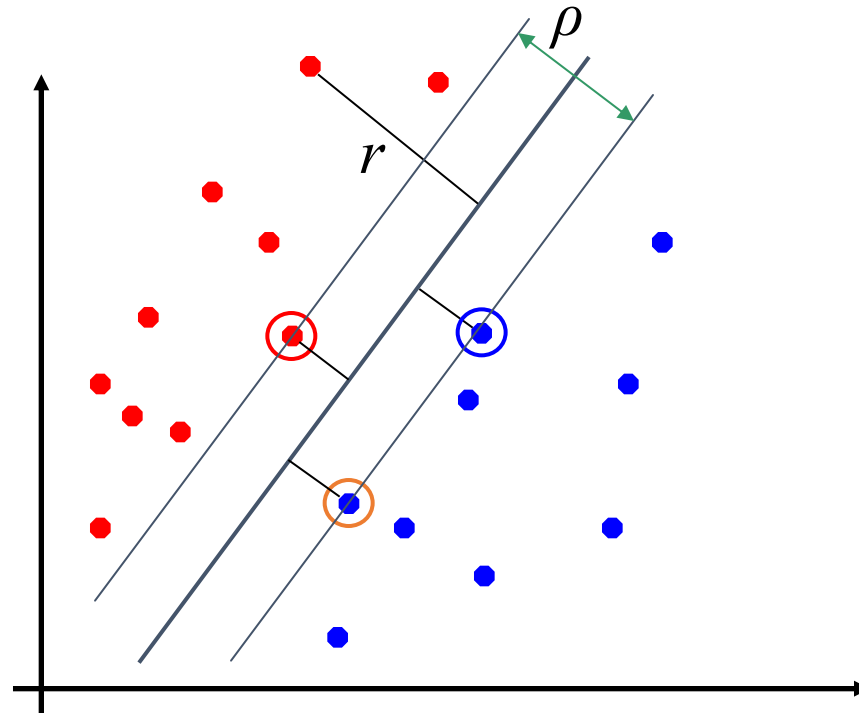
Linear Separators

- Which of the linear separators is optimal?



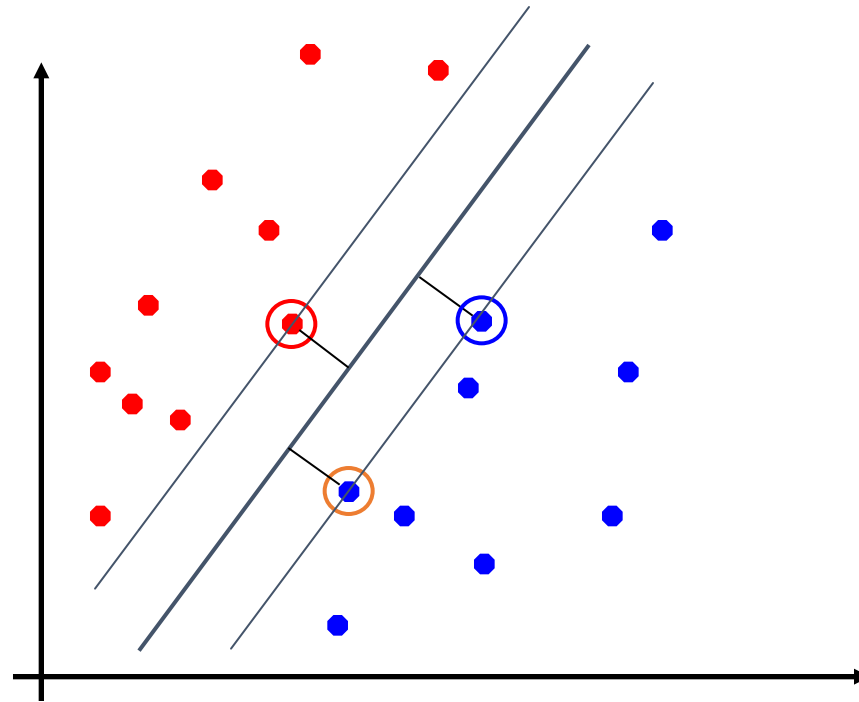
Classification Margin

- Distance from example $\mathbf{x}_i$ to the separator is $r = \frac{\mathbf{w}^T \mathbf{x}_i + b}{\|\mathbf{w}\|}$
- Examples closest to the hyperplane are *support vectors*.
- *Margin* ρ of the separator is the distance between support vectors.

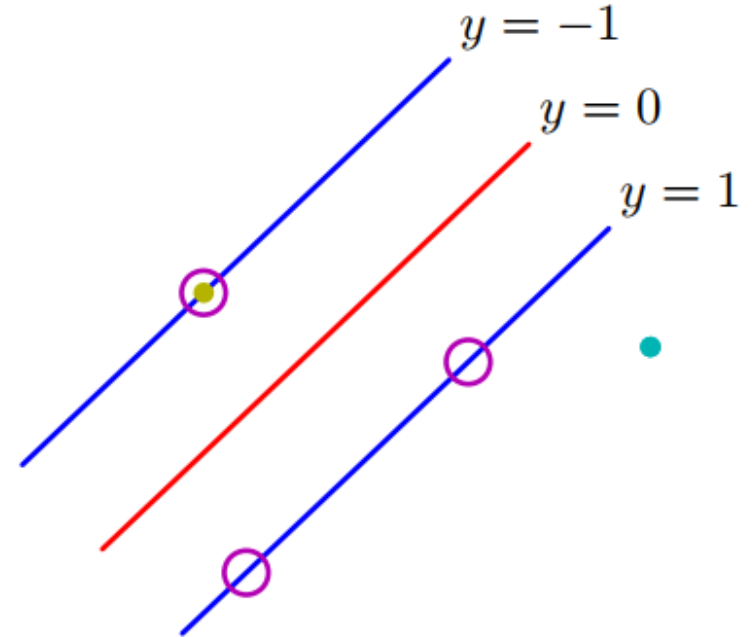
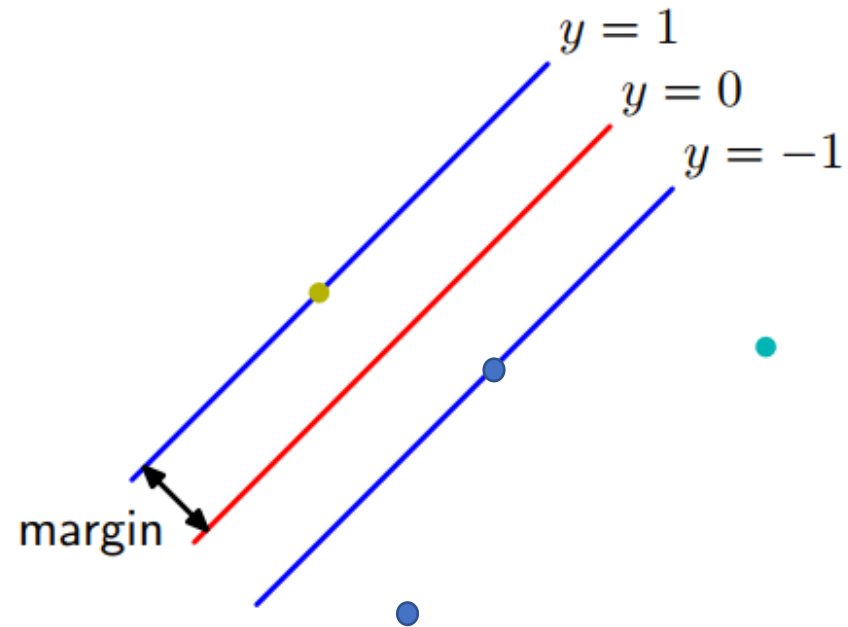


Maximum Margin Classification

- Maximizing the margin is good according to intuition and PAC theory.
- Implies that only support vectors matter; other training examples are ignorable.



Maximum Margin Classifiers



Maximum Margin Classifiers

$$y(\mathbf{x}) = \mathbf{w}^T \phi(\mathbf{x}) + b$$

N input vectors $\mathbf{x}_1, \dots, \mathbf{x}_N$
target values $t_1, \dots, t_N$
where $t_n \in \{-1, 1\}$

assume for the moment that the training data set is linearly separable

$$t_n y(\mathbf{x}_n) > 0$$

Maximum Margin Classifiers

$$y(\mathbf{x}) = \mathbf{w}^T \phi(\mathbf{x}) + b$$

- Perpendicular distance of a point $\mathbf{x}$ from this hyperplane:

$$|y(\mathbf{x})| / \|\mathbf{w}\|$$

- Furthermore, we are only interested for correct classification $t_n y(\mathbf{x}_n) > 0$
- The distance for any correctly classified point from the hyperplane

$$\frac{t_n y(\mathbf{x}_n)}{\|\mathbf{w}\|} = \frac{t_n (\mathbf{w}^T \phi(\mathbf{x}_n) + b)}{\|\mathbf{w}\|}$$

Maximum Margin Classifiers

Thus the maximum margin solution is found by solving

$$\arg \max_{\mathbf{w}, b} \left\{ \frac{1}{\|\mathbf{w}\|} \min_n \left[t_n \left(\mathbf{w}^T \phi(\mathbf{x}_n) + b \right) \right] \right\}$$

- Direct solution of this optimization problem would be very complex, and so we shall convert it into an equivalent problem that is much easier to solve.

Maximum Margin Classifiers

- We can use this freedom to set

$$t_n (\mathbf{w}^T \phi(\mathbf{x}_n) + b) = 1$$

for the point that is closest to the surface. In this case, all data points will satisfy the constraints

$$t_n (\mathbf{w}^T \phi(\mathbf{x}_n) + b) \geq 1, \quad n = 1, \dots, N. \quad \text{constraints}$$

- This implies

$$\arg \max_{\mathbf{w}, b} \left\{ \frac{1}{\|\mathbf{w}\|} \right\} \quad \text{subject to the constraints}$$

Maximum Margin Classifiers

The optimization problem then simply requires that we maximize $\|\mathbf{w}\|^{-1}$, which is equivalent to minimizing $\|\mathbf{w}\|^2$, and so we have to solve the optimization problem

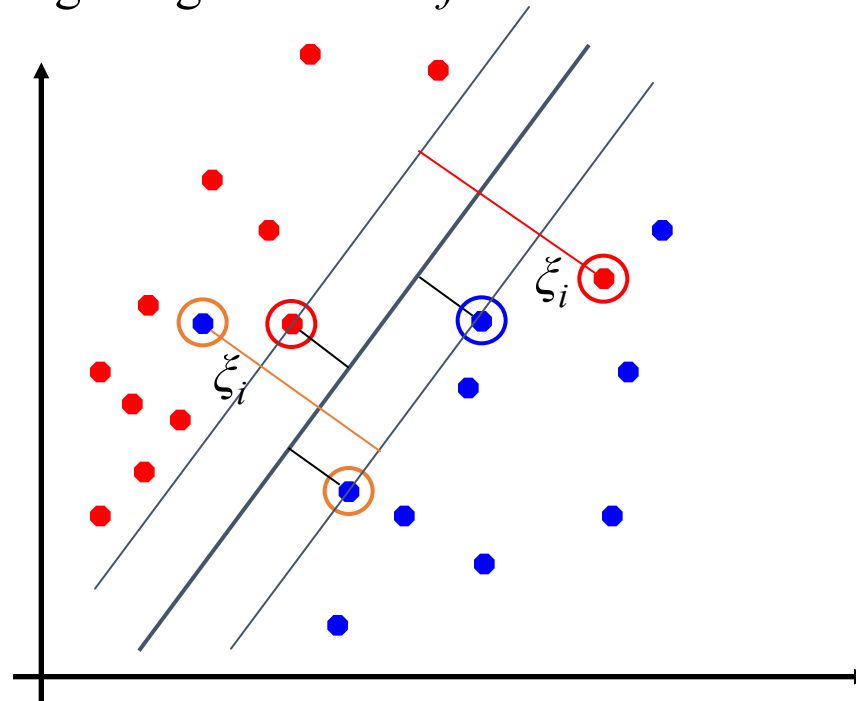
$$\arg \min_{\mathbf{w}, b} \frac{1}{2} \|\mathbf{w}\|^2$$

subject to the constraints

$$t_n (\mathbf{w}^T \phi(\mathbf{x}_n) + b) \geq 1, \quad n = 1, \dots, N.$$

Soft Margin Classification

- What if the training set is not linearly separable?
- *Slack variables* ξ_i can be added to allow misclassification of difficult or noisy examples, resulting margin called *soft*.

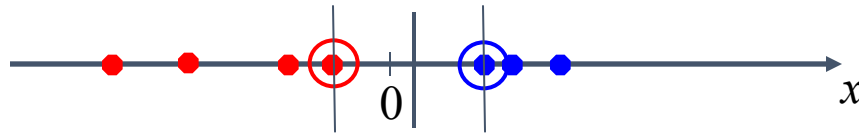


Linear SVMs: Overview

- The classifier is a *separating hyperplane*.
- Most “important” training points are support vectors; they define the hyperplane.
- Quadratic optimization algorithms can identify which training points $\mathbf{x}_i$ are support vectors.

Non-linear SVMs

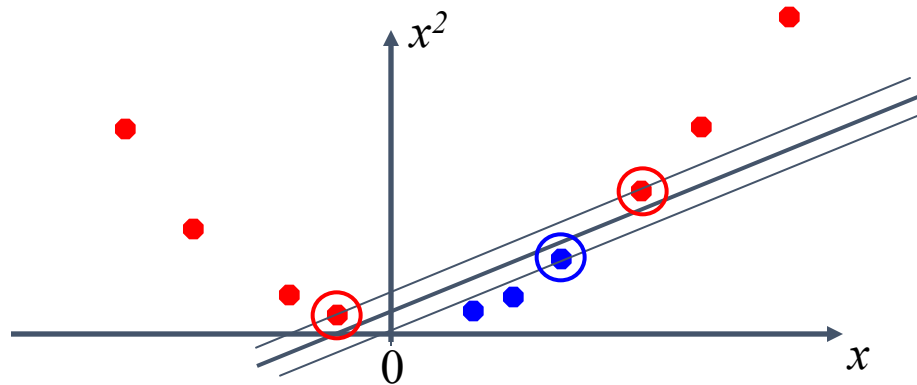
- Datasets that are linearly separable with some noise work out great:



- But what are we going to do if the dataset is just too hard?

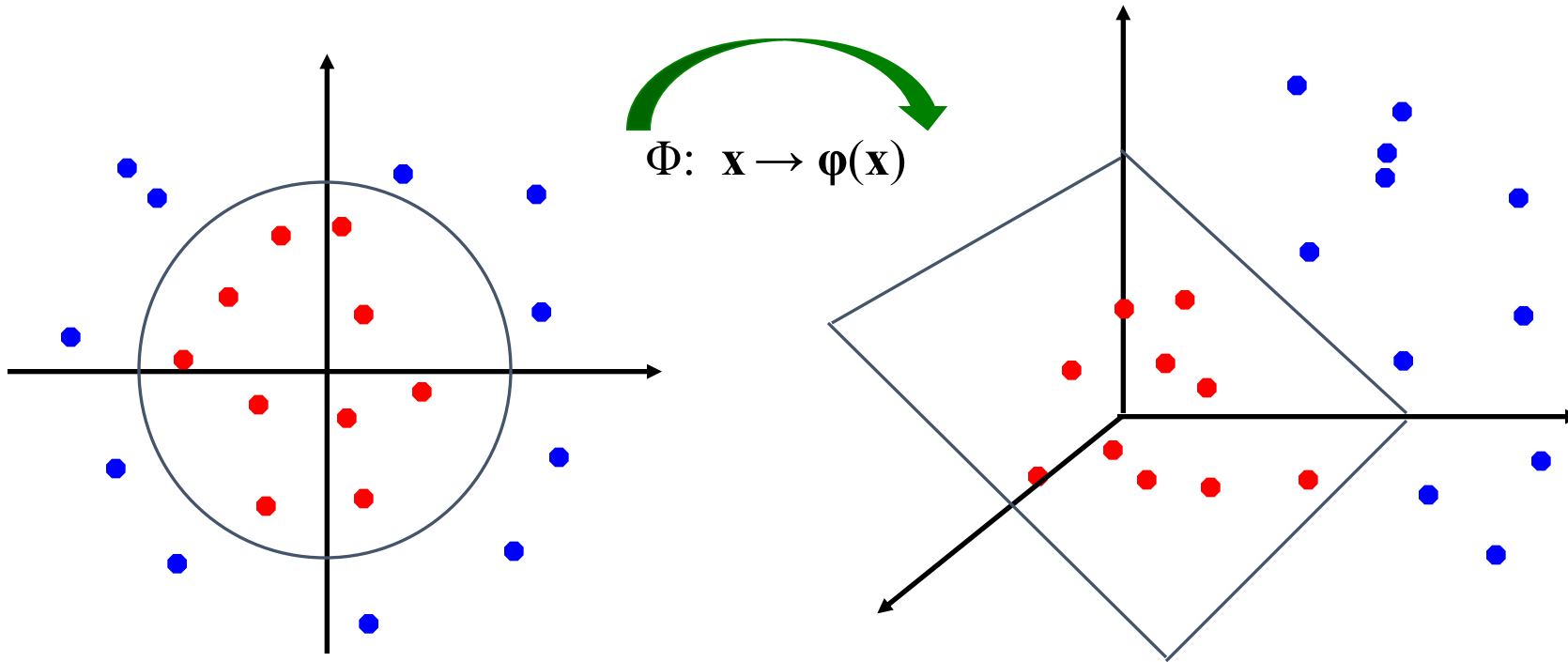


- How about... mapping data to a higher-dimensional space:



Non-linear SVMs: Feature spaces

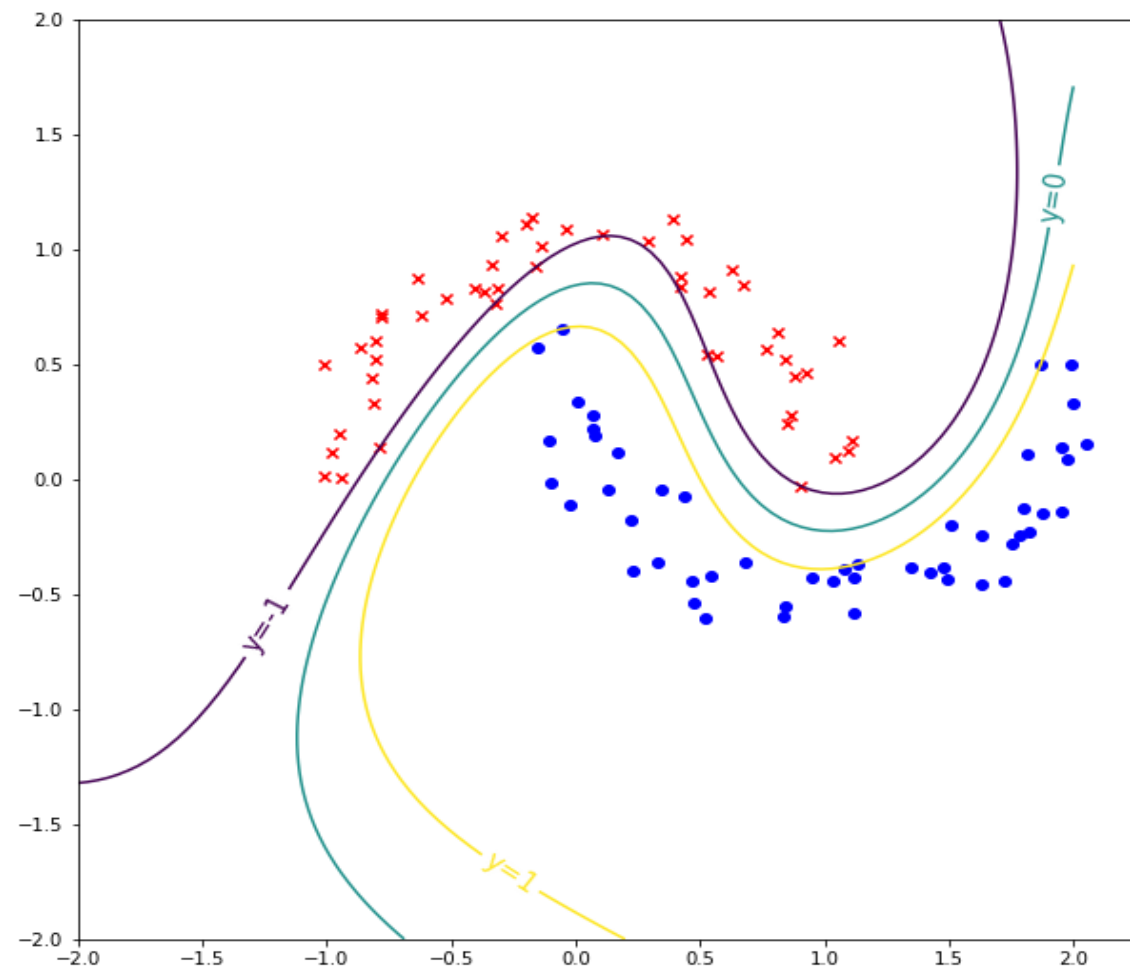
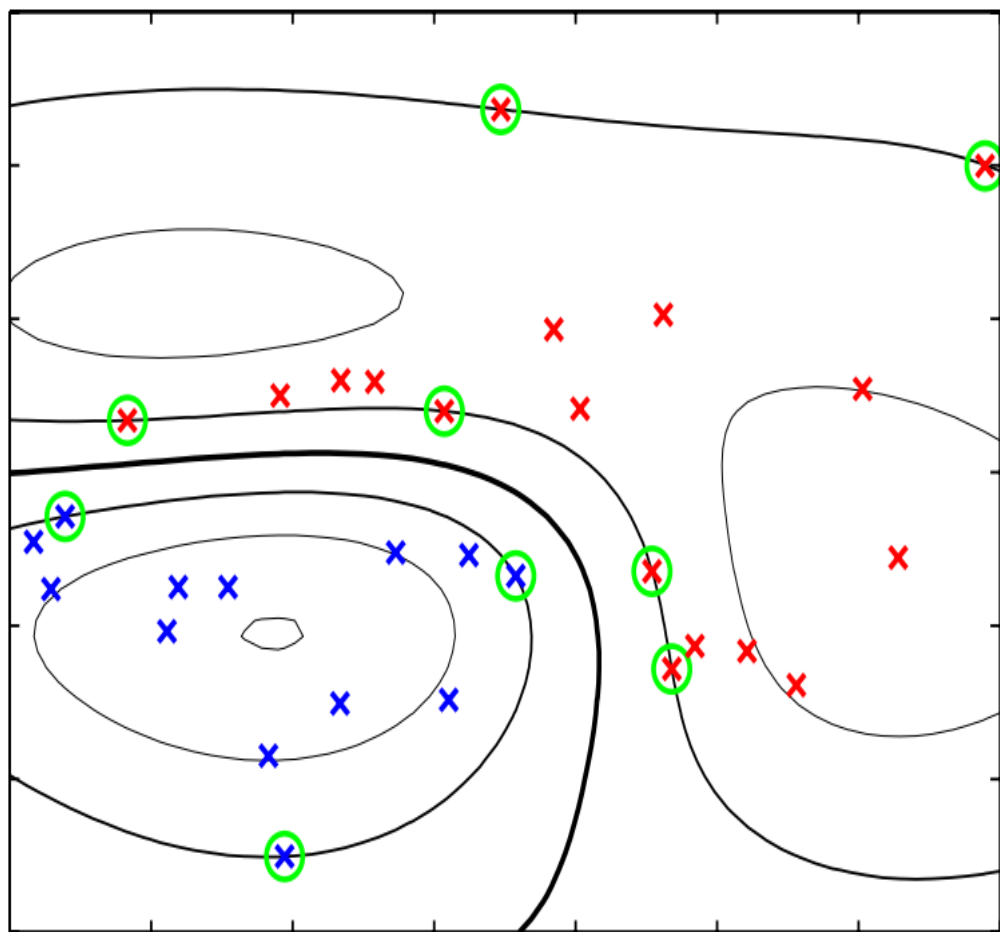
- General idea: the original feature space can always be mapped to some higher-dimensional feature space where the training set is separable:



SVM applications

- SVMs were originally proposed by Boser, Guyon and Vapnik in 1992 and gained increasing popularity in late 1990s.
- SVMs are currently among the best performers for a number of classification tasks ranging from text to genomic data.
- SVMs can be applied to complex data types beyond feature vectors (e.g. graphs, sequences, relational data) by designing kernel functions for such data.

SVM with Gaussian kernel function



CODE

- https://github.com/aruntakhur/MachineLearning_2021_IITDWD/blob/main/SupportVectorMachine.ipynb

END